nature communications

Article

https://doi.org/10.1038/s41467-026-77538-5

scTransMIL bridges patient-level disease states and single-cell transcriptomics for cancer screening and heterogeneity inference

Received: 9 July 2025

Accepted: 26 August 2026

Published online: 14 September 2026

Check for updates

Zhenchao Tang[1,2,3,7], Fang Wang[1,7] ✉, Fan Yang[1], Jiangning Song[4], Yiming Li[1], Jiale Zhou[1], Yidong Song[1], Shouzhi Chen[1], Jun Zhu[1], Linlin You[3] ✉, Calvin Yu-Chian Chen[2,5,6] ✉ & Jianhua Yao[1] ✉

Single-cell sequencing offers profound insights into tumor complexity. However, reliably linking individual cellular profiles to overall sample-level phenotypes remains computationally challenging, particularly when cell-specific annotations are absent. This disconnect limits the technology's utility in studying underlying tumor mechanisms. Here we show how scTransMIL, an artificial intelligence framework, bridges this gap. By conceptualizing a biological sample as a collection of individual cells, scTransMIL employs a transformer-based multi-instance learning approach to directly connect single-cell transcriptomic profiles with overarching sample-level labels. We demonstrate that scTransMIL accurately predicts sample-level cancer phenotypes, including the tissue-of-origin for metastatic tumors. At cellular resolution, it robustly identifies tumor-associated cell populations and maps biological progression trajectories using only minimal sample-level annotations. Furthermore, the model's attention mechanisms facilitate full-transcriptome biomarker discovery. By systematically integrating molecular and cellular scales with broader phenotypic states, our approach provides a computational tool to dissect tumor heterogeneity and advance fundamental cancer biology.

Single-cell sequencing technology is transforming cancer research and clinical diagnosis by providing high resolution into tumor heterogeneity and disease progression[1]. This high-resolution approach enables cancer detection and improved patient outcomes[2], while also revealing the complex genomic diversity that underlies variations in cancer subtypes and treatment responses. The ability to deconstruct patient-specific tumor cell populations and their molecular profiles is critical for developing personalized and targeted treatment strategies[3]. As single-cell RNA sequencing (scRNA-seq) becomes increasingly widespread and datasets grow exponentially, artificial intelligence (AI) has been widely adopted for fundamental scRNA-seq analysis tasks[4–6], including imputation[7–12], integration[13–19], and cell

[1]AI for Life Sciences Lab, Tencent, Shenzhen, China. [2]School of AI for Science, Peking University Shenzhen Graduate School, Shenzhen, China. [3]School of Intelligent Systems Engineering, Shenzhen Campus of Sun Yat-sen University, Shenzhen, China. [4]Biomedicine Discovery Institute and Department of Biochemistry and Molecular Biology, Monash University, Melbourne, Australia. [5]State Key Laboratory of Natural and Biomimetic Drugs, Peking University, Beijing, China. [6]Department of Medical Research, China Medical University Hospital, Taichung, Taiwan. [7]These authors contributed equally: Zhenchao Tang, Fang Wang. ✉e-mail: avonwanghit@gmail.com; youllin@mail.sysu.edu.cn; cy@pku.edu.cn; jianhua.yao@gmail.com

annotation[20–26]. However, AI applications that directly connect scRNA-seq data to disease states remains underdeveloped[27], hampered by significant technical challenges. Current approaches for labeling disease-associated single cells predominantly rely on copy-number variant (CNV) inference, which is limited to genetically unstable cells[28] and often confounded by the presence of normal cells in tumor samples that obscure CNV signals, especially when analyzing full transcriptome data[29,30]. For example, in Filbin et al.[31], tumor cells in pediatric gliomas were shown to be largely diploid, with malignancy driven primarily by epigenetic programs rather than large-scale CNVs, resulting in weak CNV signals. Similarly, Venteicher et al.[32] reported that malignant cells can closely resemble normal glial cells at the transcriptomic level, making CNV-based discrimination particularly challenging when CNV signals are subtle.

Publicly available cancer-related scRNA-seq datasets typically contain valuable patient-level phenotype annotations, but the lack of single-cell resolution labels restricts their utility for understanding cancer biology at the cellular level[33–35]. To address this limitation, computational methods that are capable of bridging patient-level disease states with single-cell resolution insights are needed. Against the backdrop of rapid advancements in AI, we aim to develop a data-driven deep learning model that achieves the three key objectives: accurately predicting cancer states of a given patient's sample for diagnosis, inferring cancer-relatedness at single-cell resolution to uncover underlying cellular mechanisms driving the cancer phenotype, and identifying tumor-specific genes across the full transcriptome to facilitate biomarker discovery and precision oncology.

Current approaches for linking patient-level disease states to single-cell data can be categorized into three main types, each with notable limitations. First, pseudo-bulk methods such as KIDA[36] aggregate single-cell expression profiles into sample-level feature vectors. While efficient, this averaging approach sacrifices single-cell resolution, hindering the capture of specific cell sub-populations that drive disease progression. Similarly, multi-scale integration strategies, such as DEGAS[37] and Scissor[38], attempt to align single-cell data with clinically labeled bulk data in a shared latent space. However, the emphasis on distribution alignment often over-smoothes the data, potentially obscuring subtle biological features critical for resolving tumor heterogeneity. Second, the emergence of Transformer-based foundation models, including scBERT[23], scGPT[25], and Geneformer[39], has advanced the field by learning complex gene-gene interactions through pre-training on massive single-cell atlases. While these models achieve strong performance in cell-level tasks like cell type annotation, they lack a dedicated mechanism for aggregating features from thousands of cells into patient-level diagnostic predictions. This is particularly challenging in weakly supervised clinical scenarios where cell-level ground truth labels are unavailable. Third, weakly supervised Multi-Instance Learning (MIL) has proven effective in handling such label ambiguity. In the single-cell domain, scIDST[40] attempted to utilize instance-based MIL, but this approach often introduces noise by directly assigning bag labels to all constituent cells. Conversely, bag-based methods like MultiMIL[41] are more robust at the aggregation level but often struggle with the extreme class imbalance found in cancer samples, where a small number of malignant cells may be masked by a vast majority of normal cells.

MIL, a weakly supervised learning approach widely used in whole-slide image analysis, has demonstrated its effectiveness in handling weakly labeled biomedical data[42–47]. It treats patient's samples as "bags", and individual cells within samples as "instances". However, conventional MIL approaches suffer from critical shortcomings: Instance-based MIL inaccurately assigns the bag's label to all cells, introducing noise, while bag-based MIL optimizes only at the sample level, often over-fitting to easily identifiable positive cells while neglecting harder-to-classify yet disease-relevant instances[48].

In this work, to address these limitations, we introduce scTransMIL, a scRNA-seq Transformer-based Multi-Instance Learning framework designed to bridge patient-level cancer states and single-cell data. scTransMIL systematically overcomes existing technical bottlenecks through: (1) a full transcriptome-wide Transformer encoder that utilizes rank value encoding to capture contextual gene relationships while mitigating batch effects; (2) self-attention mechanisms enabling direct gene-level disease state-specific interpretability; (3) a gated attention mechanism to filter noise during bag-level aggregation; (4) contrastive learning to enhance the discrimination between positive and negative cells, improving the identification of challenging yet biologically meaningful cell populations; and (5) an adaptive clustering loss that enhances instance-level discrimination even under weak supervision. Trained on large-scale pan-cancer datasets, scTransMIL demonstrates strong performance in predicting cancer states for unseen patient's samples and inferring cancer-relatedness at single-cell resolution compared to existing methods. Moreover, scTransMIL provides gene-level insights into tumor biology and exhibits robust generalization across independent cancer datasets. These capabilities position scTransMIL as a useful tool for advancing cancer diagnosis, dissecting tumor heterogeneity, and accelerating biomarker discovery.

Results

Overall model architecture of scTransMIL

In the scTransMIL framework, MIL treats each patient's sample as a "bag", and individual cells within the sample as "instances". While cancer states (bag labels) are available at the patient level, the phenotype of each cell (instance label) remains unknown. According to the theoretical assumption of MIL, a bag is labeled as normal only if all its instances are normal cells; conversely, any bag containing at least one tumor-related cell is labeled as a tumor bag (Fig. 1a). Taking bags containing multiple instance cells as input, the MIL framework relies solely on bag labels for supervised learning and thus predicts labels for query bags (Fig. 1b). Specifically, scTransMIL operates in two stages (Fig. 1c). In Stage 1 of scTransMIL, an Instance encoder generates initial embeddings for each cell in the query bag, a Gated encoder outputs gated instance embeddings and instance-level tumor scores that can be used to indicate the malignancy of each cell, and a Bag aggregator integrates these gated instance embeddings to output a bag embedding, which is further classified as tumor or normal. If classified as a tumor bag, it enters Stage 2, which employs a multi-classifier to predict the specific cancer type using the bag embedding from Stage 1.

To ensure accuracy and robustness, scTransMIL incorporates several critical designs. For the Instance encoder (Supplementary Fig. 1), the transcriptomic profile of each cell is encoded into a ranked gene sequence, for which potential batch effect is expected to be removed since gene ranking is more robust than expression values. The ranked gene sequence of each cell is fed into a 6-layer Transformer to generate initial cell embeddings. The Transformer architecture is particularly effective at processing rank-encoded sequences because its self-attention mechanism captures complex gene-gene co-expression relationships rather than relying on absolute expression magnitudes. By focusing on the relative contextual hierarchy of the transcriptome, scTransMIL identifies robust biological patterns that are less sensitive to technical noise and batch-specific variations.

In the Gated encoder (Fig. 1d), a learnable gate filters bag label-irrelevant information from the initial cell embeddings, generating gated embeddings, which are then linearly projected to produce instance-level tumor scores. In the Bag aggregator (Fig. 1d), all gated embeddings and tumor scores in the bag are weighted and summed into a bag embedding, which is fed into the classifier to predict the bag label. During model training, the bag loss and adaptive clustering loss jointly guide the parameters update. The bag loss only requires the bag label and uses cross-entropy to update the entire scTransMIL.

https://doi.org/10.1038/s41467-026-77538-5

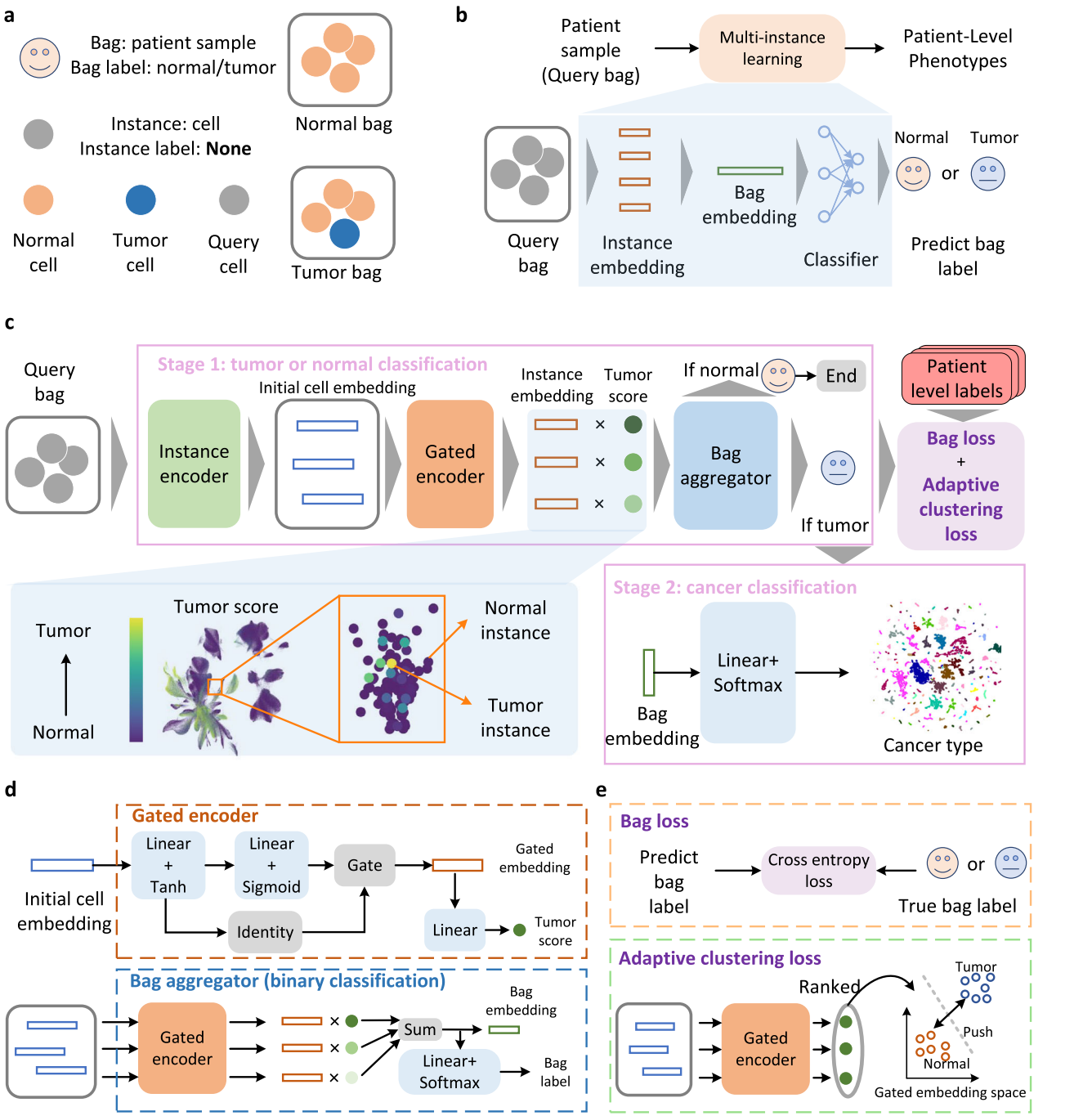


Fig. 1 | The overall framework of scTransMIL. a Schematic of the multi-instance learning (MIL) data format. Each sample is treated as a bag containing multiple single-cell instances. Bag-level labels (e.g., tumor or normal) are available, while instance-level labels remain unknown. **b** Schematic of the MIL paradigm. For a query bag, single-cell instances are encoded into embeddings, aggregated into a bag embedding, and passed through a classifier to predict patient's sample-level phenotypes. **c** Two-stage prediction process of scTransMIL. Stage 1 performs binary classification to distinguish tumor from normal samples, while simultaneously assigning tumor scores to individual cells. For tumor samples, Stage 2 further classifies cancer types using the bag embedding. In the main pipeline (stage 1), scTransMIL utilizes a transformer-based instance encoder to transform transcriptomics data into initial cell embeddings (refer to Supplementary Fig. 1). The gated encoder updates the initial cell embeddings into instance embeddings and tumor scores. Finally, the bag aggregator aggregates the instance embeddings into a bag embedding. **d** Architectures of the gated encoder and the bag aggregator. The gated encoder generates tumor-relevant gated embeddings and assigns tumor scores to individual cells by applying a gating mechanism. The bag aggregator combines gated embeddings from all instances within a bag to produce a bag-level representation, which is then used for bag label prediction via a linear classifier and softmax function. **e** Loss functions in scTransMIL. The bag loss applies cross-entropy loss based on the predicted bag label and the true bag-level disease state. The adaptive clustering loss ranks instances by tumor scores and encourages separation within the gated embedding space, enhancing instance-level discrimination while mitigating overfitting in weakly supervised learning.

Additionally, the adaptive clustering loss sorts gated embeddings by tumor scores to maximize separation between the gated embeddings at the extremes of the sorted sequence, thereby mitigating over-fitting in positive instance identification (Fig. 1e).

Comprehensive evaluation of tumor prediction across pan-cancer data

We first evaluated the performance of scTransMIL against several baseline methods, including KIDA[36], DEGAS[37], scIDST[40], MultiMIL[41] and two variants of scTransMIL (scTransMIL(PCA) and scTransMIL(FM)) on a pan-cancer dataset[49]. To make fair comparison with MultiMIL[41], which was originally designed for multi-omic data, we adopted its MIL module for single-omic data analysis by generating input embeddings using scVI. Additionally, we also developed two scTransMIL variants: scTransMIL(PCA) utilizing principal component analysis (PCA)-reduced cell embeddings and scTransMIL(FM) incorporating cell embeddings derived from a foundation model (fully pretrained scGPT[25]), both serving as alternatives to our Instance encoder's initial embeddings.

Our evaluation framework employed a comprehensive pan-cancer dataset[49] (comprising 4 million cells from 1563 samples in 999 donors across 30 cancer types, aggregated from 104 public datasets) to rigorously assess the model performance using Accuracy (Acc), Area Under the Receiver Operating Characteristic Curve (AUC), and F1 Score (F1) metrics. The pan-cancer dataset's broad diversity across cancer types provided an ideal testbed for evaluating the model's generalization capabilities across different tumor profiles. We implemented strict separation of data by randomly splitting the dataset into 80% training, 10% validation, and 10% test sets based on donors, ensuring that donors in the test and validation sets were completely unseen during training. This design allows for a rigorous evaluation of scTransMIL's ability to predict cancer states from single-cell data, highlighting its robust performance across diverse cancer types and new patient's samples.

To evaluate the impact of training data size on model performance, we generated three stratified subsets of different sizes, 50, 75, and 100% of the samples randomly drawn from the aforementioned 80% training donor pool. We trained scTransMIL and the comparison methods on each training subset separately while maintaining a fixed test set with samples derived from the remaining 10% of unseen donors for unbiased evaluation. As illustrated in Fig. 2a, as the scale of training data increases, scTransMIL achieves better performance.

To ensure a standardized assessment of computational efficiency, we measured the inference time of scTransMIL and all baseline methods required for processing a single bag on an identical hardware server (CPU: 10 cores, 2.5 GHz/core, GPU: NVIDIA TESLA V100) (Fig. 2b). For variants requiring external feature extraction, such as scTransMIL(PCA) and scTransMIL(FM), the reported times incorporate the overhead for PCA dimensionality reduction and scGPT foundation model forward-pass computation, respectively. Our results indicate that scTransMIL (245.16 s) provides a favorable trade-off between the high-resolution features captured by its transformer architecture and the time-intensive processing of methods like DEGAS (720.00 s), making it suitable for large-scale atlas analysis.

To further validate the importance and effectiveness of scTransMIL's key design elements, we conducted ablation experiments through removing the adaptive clustering loss (Ablation1), gating mechanism (Ablation2), or replacing the attention pooling with max pooling (Ablation3) (Supplementary Fig. 1). As shown in Fig. 2c, removal of any of these key components resulted in a decline in model performance. Notably, models using full transcriptome data as input consistently outperformed those relying on PCA-reduced embeddings, highlighting the value of preserving full transcriptome information for cancer phenotype inference.

Furthermore, the gated encoder is designed to suppress cell-level variability that is irrelevant to bag-level labels while preserving discriminative signals associated with disease states. To quantify this effect, we evaluated cell embeddings before and after the gated encoder using supervised clustering metrics, including normalized mutual information (NMI), adjusted Rand index (ARI), adjusted mutual information (AMI), and homogeneity score (HOM), based on cell labels provided in the pan-cancer dataset[49]. As shown in Supplementary Fig. 2, embeddings after the gated encoder consistently achieve higher scores across both binary and multi-class classification settings, indicating that the gating mechanism selectively filters label-irrelevant information while enhancing label-discriminative structure. These results demonstrate that the gated encoder refines instance representations in a task-aware manner rather than simply compressing information.

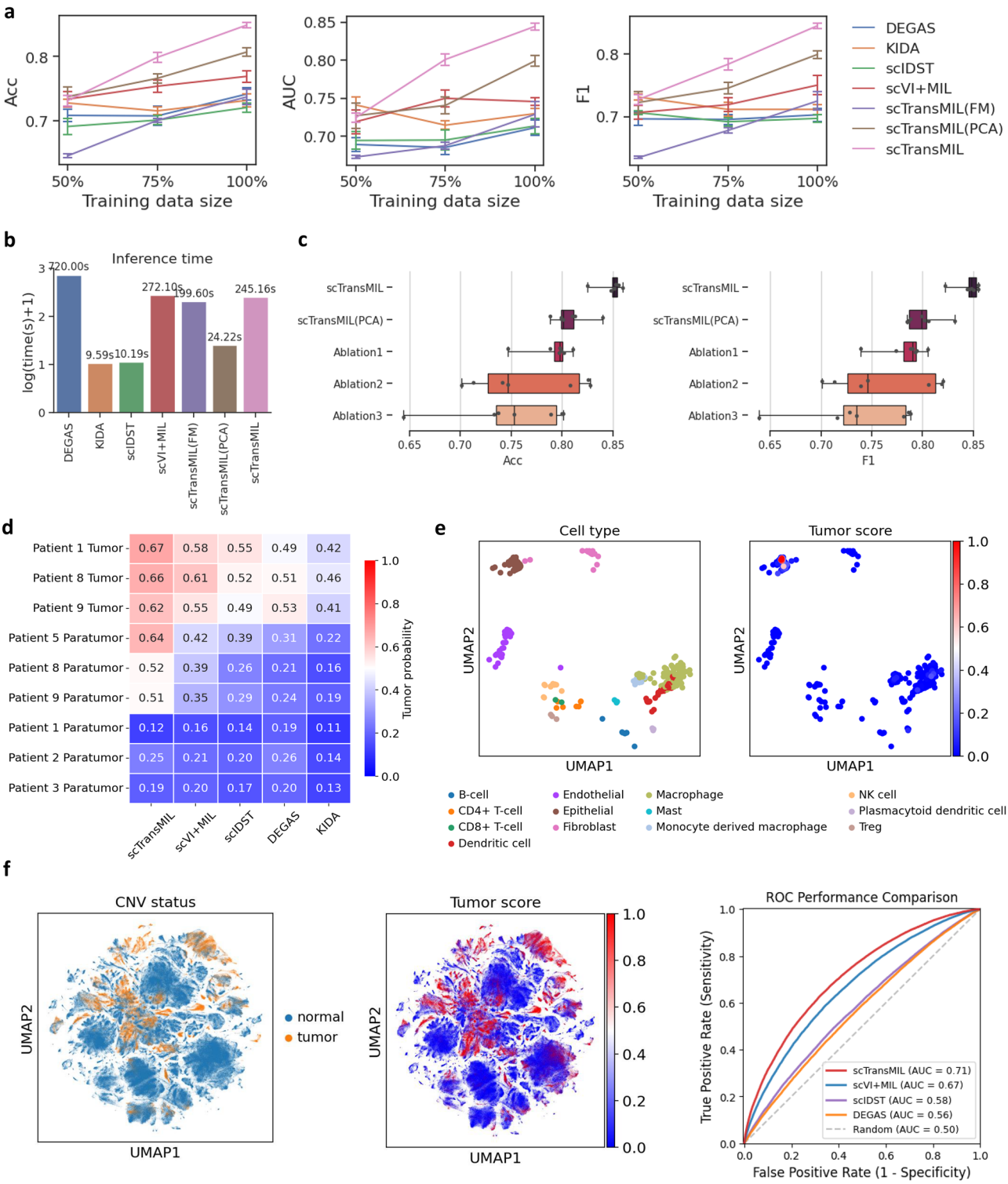
To rigorously assess model robustness under challenging real-world conditions, we compared performance across two distinct labeling strategies ("Methods" subsection "Training") on the pan-cancer dataset[49]: the Sample setting (default setting) and the Donor setting (Supplementary Fig. 3a). While the Sample setting utilizes sub-sample-level label ("Methods" subsection "Data preprocessing"), the Donor setting reflects a common clinical scenario where sub-sample-specific annotations are unavailable, and all sub-samples from a cancer-diagnosed donor are uniformly labeled as tumor-positive. Note that we perform inference on sub-samples, and then aggregate the results back to the full sample for evaluation ("Methods" subsection "Inference"). This donor-level assignment introduces significant label noise, as non-malignant tissues (e.g., paratumor or normal adjacent samples) are incorrectly categorized as tumor-positive (Supplementary Fig. 3a). Comparative analysis revealed that while all methods experienced a performance decline under these noisy conditions, MIL-based approaches, and scTransMIL in particular, maintained stability and resistance to label noise across all key metrics (Supplementary Fig. 3b). These results provide empirical validation that the scTransMIL framework is well suited for bridging coarse patient-level disease states with high-resolution single-cell data, maintaining high-fidelity predictions even in the presence of the ambiguous labeling commonly encountered in large-scale clinical cohorts.

This resilience is further exemplified in our thyroid cancer detection analysis. Early-stage cancer detection is particularly challenging because malignant signatures are often diluted by a dominant normal microenvironment and may involve transitional cell states that differ significantly from late-stage tumors. As a validation of the potential for cancer detection, we validated the model on a biologically complex thyroid cancer dataset (GSE184362[50]), in which localized tumor samples represented early-stage disease and paratumor samples from multiple patients were used as normal controls as described in the original study.

We first benchmarked scTransMIL on localized thyroid tumor samples (Patients 1, 8, and 9) against paratumor controls. As shown in Fig. 2d, baseline methods exhibited limited sensitivity in this localized context. Specifically, pseudo-bulk-based methods like KIDA[36] consistently produced prediction scores below 0.5, as the averaging of gene expression across thousands of cells effectively 'washed out' the rare signals from the sparse malignant population. Similarly, MIL-based frameworks such as DEGAS[37] and scIDST[40] struggled to differentiate these early signals from the technical noise of the normal background.

The most compelling evidence of scTransMIL's sensitivity emerged from the analysis of a paratumor sample from Patient 5. As reported by the original study[50], although this sample exhibits normal tissue architecture in pathological sections, single-cell transcriptomic analysis reveals a distinct epithelial subpopulation (cluster c02) that has begun to express cancer-associated genes. These cells correspond to thyroid follicular epithelial cells (thyrocytes) in a precancerous

https://doi.org/10.1038/s41467-026-77538-5



state, representing a biologically realistic early-stage lesion with very low tumor cell abundance. However, scTransMIL flagged this sample as positive with a prediction score of 0.64, while all other methods, including the scVI+MIL baseline, classified it as negative (Fig. 2d).

To investigate the source of this positive prediction, we performed a cell-level interpretability analysis by projecting scTransMIL's inferred tumor scores onto the single-cell UMAP landscape of Patient 5 (Fig. 2e). We observed that high tumor scores were not randomly distributed but were specifically and consistently concentrated within the epithelial cell population, while immune and stromal cells show uniformly low scores. This localization is biologically coherent with the original annotation of the dataset and demonstrates that scTransMIL can selectively highlight rare precancerous epithelial cells embedded within an otherwise normal cellular background, rather than being driven by global cell composition or random subsampling effects.

To rigorously evaluate scTransMIL's ability to infer disease-relatedness at the single-cell resolution, we performed comprehensive validation of its disease-relatedness predictions across the entire 4

Fig. 2 | Benchmarking and robustness evaluation of scTransMIL for tumor prediction on pan-cancer dataset. a Sample-level diagnostic performance (Tumor vs. Normal) of scTransMIL, its variants, and baseline methods (DEGAS, KIDA, scIDST, scVI + MIL) across different training data sizes (50, 75, 100% of samples derived from training donor pool) using Accuracy, AUC, and F1 score as evaluation metrics ($n = 7$). scTransMIL consistently outperforms baselines, with greater performance gains as training data increases. Data are presented as mean ± s.d. across $n = 7$ independent random data splits; the center denotes the mean. **b** Computational efficiency comparison. Bar plot showing the log-transformed inference time (in seconds) for scTransMIL, its variants, and four baseline methods measured on the same standardized server. For scTransMIL(PCA) and scTransMIL(FM), the times include PCA pre-processing and scGPT embedding generation, respectively, ensuring a comprehensive end-to-end performance assessment. **c** Ablation study assessing the impact of key components in scTransMIL (adaptive clustering loss, gating mechanism, and attention pooling). Removal of any component leads to decreased performance in both Accuracy and F1 score, confirming their contributions. Ablation1: removing the adaptive clustering loss. Ablation2: removing the gating mechanism. Ablation3: replacing the attention pooling with max pooling. Please refer to Supplementary Fig. 2 for the three ablation settings. $n = 7$ independent random data splits. In the box plots, the center line denotes the median, the box bounds denote the 25th and 75th percentiles (interquartile range), and the whiskers extend to the minimum and maximum values. **d** Case study of early-stage cancer sample prediction. In addition to correctly identifying three cancer samples, scTransMIL successfully identified the paratumor sample from Patient 5 as positive. Notably, while this sample exhibits normal tissue structure in pathological images, it has already begun to express cancer-associated genes at the molecular level. **e** Visualization of the single-cell landscape for the Patient 5 paratumor sample. Given that this sample contains thyroid follicular epithelial cells in a precancerous state, the regions with high tumor scores inferred by scTransMIL are consistently concentrated within the epithelial cells. **f** Single-cell resolution UMAP visualization and statistical evaluation. The panels display the CNV-based annotations as a computational reference (left), the tumor scores inferred by scTransMIL (middle), and the ROC performance comparison at single-cell resolution (right). Source data are provided as a Source Data file.

million-cell pan-cancer atlas[49]. As shown in Fig. 2f, we visualized single-cell tumor scores predicted by scTransMIL. We observed similar distribution between scTransMIL's predictions and CNV-derived annotations (CNV status). In addition, to statistically evaluate the reliability of our cell-level predictions, we performed ROC analysis across the pan-cancer cohort using CNV-based annotations as the primary benchmark (Fig. 2f, right panel). scTransMIL achieved a higher AUC of 0.71, outperforming baseline methods such as scVI+MIL (AUC = 0.67), scIDST (AUC = 0.58), and DEGAS (AUC = 0.56). We also evaluated Acc and F1 at single-cell resolution (Supplementary Fig. 2b).

Prediction of cancer types across primary and metastatic samples

Building upon its accurate tumor/normal classification, scTransMIL's multi-classifier module shows a strong capability in determining specific cancer types—a feature with particular clinical relevance and value for challenging clinical applications like Cancer of Unknown Primary (CUP)[51]. CUP presents a significant diagnostic dilemma, where metastatic lesions are identified without detectable primary tumors. This diagnostic uncertainty poses significant clinical challenges, as accurately identifying the tumor origin is critical for determining optimal treatment strategies[52] yet remains particularly difficult in such cases. Leveraging the comprehensive cancer type annotations provided in our pan-cancer dataset (30 distinct types, Supplementary Fig. 4a and Supplementary Data 1), scTransMIL achieved better classification accuracy for both primary and metastatic tumors in held-out test samples (Fig. 3a, b). The model's robust performance on metastatic specimens is especially noteworthy, as it directly addresses a critical unmet need in clinical diagnosis and management of CUP.

Furthermore, to rigorously evaluate scTransMIL's generalizability, we extra tested its performance on a breast cancer (BRCA) dataset[53] from an independent research, which contains 400,000 cells from 37 samples corresponding to 37 patients (Supplementary Fig. 4b). In this dataset, all samples were malignant breast cancer and accordingly all bags were assigned the BRCA label. As a result, scTransMIL demonstrated robust classification accuracy with almost all primary samples correctly identified as BRCA, and with only a small fraction of metastatic samples misclassified as other cancer types (Fig. 3c), presumably due to the complexity of unseen metastatic samples.

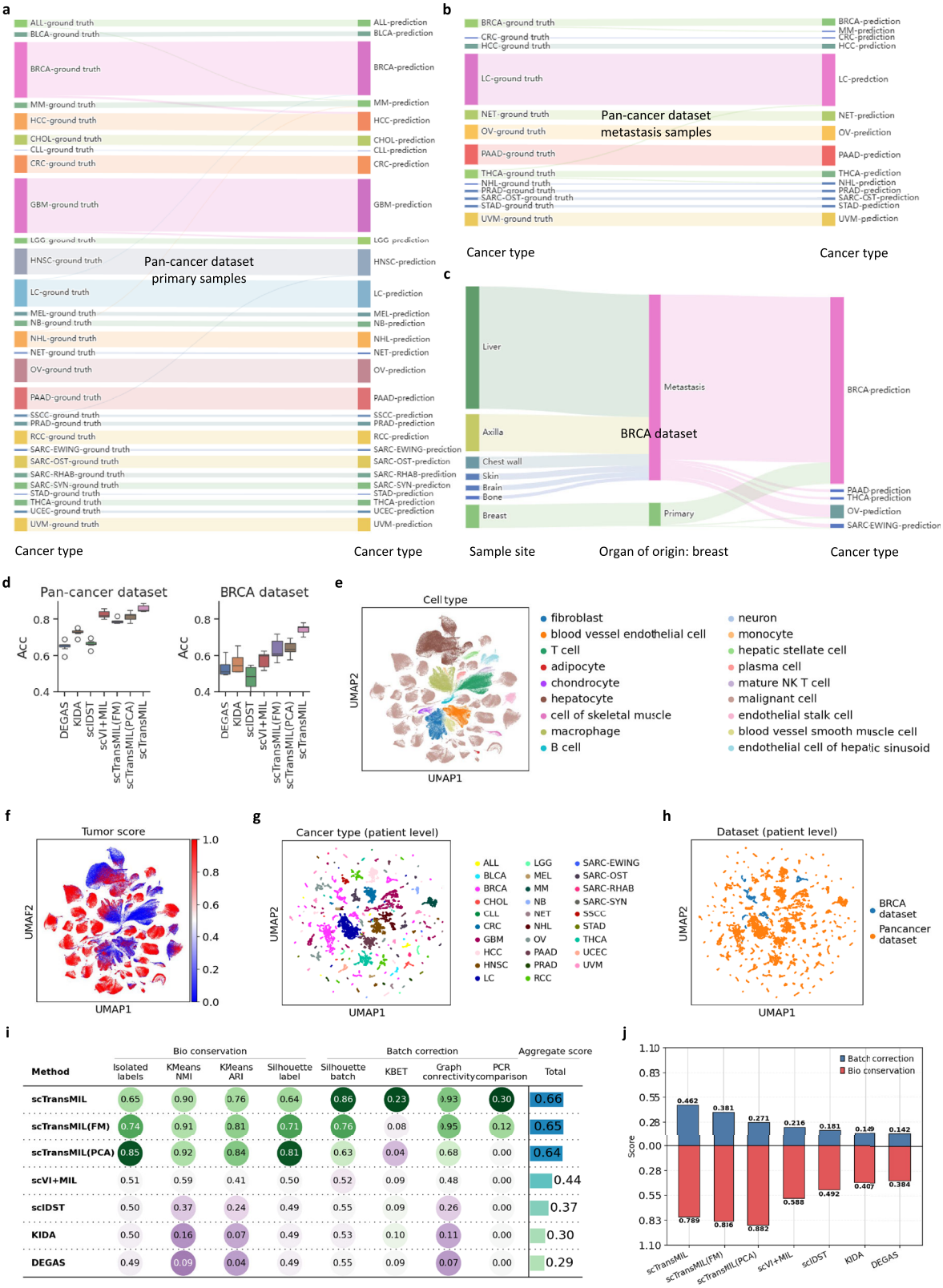
To investigate the factors that may underlie the misclassification of certain metastatic BRCA samples, we further analyzed their cellular composition, metastatic sites, and receptor status (Supplementary Fig. 5). Our analysis suggests that these mispredictions are likely associated with specific biological contexts. For instance, Thyroid Carcinoma (THCA) misclassifications appear to be linked to "signal dilution" from high lymphoid infiltration[54] (Supplementary Fig. 5a, b), while Ewing Sarcoma (SARC-EWING) mispredictions are potentially associated with bone-metastatic stromal signals[55] (Supplementary Fig. 5a–c). Notably, samples with a triple-negative status were frequently misclassified as Pancreatic Adenocarcinoma (PAAD) or Ovarian Cancer (OV), which may reflect a degree of molecular similarity between these high-grade, aggressive disease states[56] (Supplementary Fig. 5a, d).

Our comparative analysis revealed scTransMIL's strong generalization capability through benchmarking experiments against established methods on both the pan-cancer test set and the extra BRCA dataset (Fig. 3d). While all the comparison methods showed significant performance degradation when applied to the BRCA dataset, scTransMIL maintained accurate and robust classification performance, demonstrating strong generalization capability. Beyond cancer type prediction, scTransMIL could also infer the tumor scores at the single-cell resolution to the BRCA data. As shown in Fig. 3e, f, cells with high tumor scores assigned by scTransMIL were enriched in malignant cell populations in BRCA data, demonstrating robust cross-dataset generalization at single-cell resolution. Evaluation results at single-cell resolution demonstrate that scTransMIL outperforms the comparison methods (Supplementary Fig. 2c).

To evaluate scTransMIL's generalization capability across independent clinical cohorts, we analyzed the latent space representations of the pan-cancer datasets and the BRCA dataset by extracting and visualizing bag-level embeddings. The UMAP visualizations and subsequent quantitative benchmarking revealed several key findings: (1) Preservation of biological identity. The learned representations formed distinct, cancer-type-specific clusters in both the pan-cancer and BRCA datasets, demonstrating that the model successfully preserves biologically meaningful patterns across diverse lineages (Fig. 3g). (2) Bridging domain gaps in a unified space. Rather than merely removing batch effects, scTransMIL projects independent datasets into a unified latent space, effectively bridging the domain gaps between the pan-cancer and independent BRCA cohorts (Fig. 3h). This alignment ensures that samples from disparate sources are organized within a consistent biological context, facilitating robust cross-dataset classification. (3) Quantitative validation of dataset integration. To move beyond qualitative assessment, we performed evaluation using the scIB[57] (a single-cell integration benchmark framework). We use sample embedding as cell embedding, cancer type as cell type, dataset ID (2 datasets: Pan-cancer and BRCA) as batch ID.

The scIB framework comprehensively evaluates latent representations across two main dimensions: Biological Conservation and Batch Correction. Biological Conservation assesses how well true biological signals are preserved, measured by four metrics: Isolated labels (evaluating the separation of rare populations), KMeans NMI

https://doi.org/10.1038/s41467-026-77538-5



(Normalized Mutual Information) and KMeans ARI (Adjusted Rand Index) (both assessing the agreement between unsupervised clustering and true biological labels), and Silhouette label (measuring the tightness and separation of biological clusters). Conversely, Batch Correction evaluates the removal of technical noise, utilizing four metrics: Silhouette batch (assessing the mixing of batches within the same cancer type), KBET (k-nearest neighbor batch-effect test, evaluating local batch distribution in sample neighborhoods), Graph connectivity (measuring whether samples of the same type from different batches are connected), and PCR comparison (Principal Component Regression, measuring the reduction of variance contributed by batch effects). As shown in Fig. 3i, j, scTransMIL achieved a higher Batch

Fig. 3 | Generalization performance of scTransMIL in cancer type prediction. **a**, **b** Sankey diagrams showing cancer type prediction results by scTransMIL on the pan-cancer dataset. **a** Primary tumor samples; **b** Metastatic samples. Ground truth cancer types are accurately matched with predicted types, including challenging metastatic cases relevant to cancers of unknown primary (CUP). **c** Sankey diagrams of the cancer type prediction on the external validation BRCA dataset containing primary and metastatic breast cancer samples. **d** Comparison of cancer type prediction accuracy between scTransMIL, its variants, and baseline methods on both the pan-cancer and BRCA datasets. There are $n = 7$ repeats with different random seeds for data splitting. In the box plots, the center line denotes the median, the box bounds denote the 25th and 75th percentiles (interquartile range), and the whiskers extend to the minimum and maximum values. UMAP visualization of single-cell embeddings for the BRCA dataset, colored by cell type annotations (**e**) and tumor scores inferred by scTransMIL at single-cell resolution (**f**). UMAP visualization of bag embeddings for the pan-cancer and BRCA datasets, colored by cancer types (**g**) and dataset sources (**h**). **i** Quantitative evaluation of latent sample embeddings. A comprehensive benchmarking using the scIB framework to evaluate dataset integration sub-metrics across the Pan-cancer and BRCA datasets. **j** A comparison of different methods for dataset integration, focusing on Batch Correction and Bio Conservation. Source data are provided as a Source Data file.

Correction score and a Total Aggregate score, significantly outperforming variants scTransMIL(FM), scTransMIL(PCA), and comparison methods. This enhanced integration capability is attributed to our disease-specific Transformer architecture, which was pre-trained from scratch on extensive clinical datasets. By prioritizing robust diagnostic signatures over dataset-specific technical variations, scTransMIL learns biologically informed representations that enable accurate cross-dataset generalization. Batch correction results indicate that scTransMIL is capable of reducing the domain gap between two datasets, while Bio conservation results demonstrate that scTransMIL preserves the discriminative features distinguishing different cancer types. Furthermore, we observed that compared to scTransMIL (PCA) and scTransMIL (FM), scTransMIL achieves superior batch correction but exhibits poorer biological conservation. This discrepancy arises because scTransMIL's instance encoder discards specific gene expression signals; this approach effectively mitigates batch effects stemming from the specific numerical values of expression levels, thereby leading to improved batch correction. Concurrently, however, the absence of fine-grained expression information attenuates the biological heterogeneity among samples, resulting in diminished biological conservation.

Identification of tumor-specific genes in ALL using scTransMIL

scTransMIL's attention mechanism leverages the rich full transcriptome context learned from the pan-cancer dataset to provide comprehensive gene-level biological insights, enabling systematic identification of cell type-specific genes. Our analysis of acute lymphoblastic leukemia (ALL) patient's samples[58], which form distinct clusters in the embedding space of the pan-cancer dataset (Fig. 4a), revealed similar distribution between scTransMIL's single-cell tumor scores and orthogonal corrected CNV-based malignancy annotations (Fig. 4b, c and Supplementary Fig. 2d), with a prominent malignant B-cell population (Fig. 4d).

scTransMIL also assigned relatively high tumor scores to a subset of monocyte-derived macrophages in ALL patients (Fig. 4c, d). Further analysis revealed that these cells express high levels of established tumor-associated macrophage (TAM) markers, such as *CD163*, *IL10*, and *LGALS*[59,60] (Supplementary Fig. 6). We observe that this cell cluster exhibits a normal CNV state, and the TAMs are also normal cells; this suggests that scTransMIL may have captured tumor-associated transcriptional features within the microenvironment, thereby assigning a high tumor score to this cell cluster.

Additionally, to evaluate scTransMIL's resolution in complex clinical subtypes, we further investigated B-cell populations across different ALL subtypes (Fig. 4e, left). We observed that the CNV-based reference labels varied by subtype; while B-cells in ETV6-RUNX1 samples were designated as malignant, those in two High Hyperdiploid (HHD) ALL samples, HHD_1 and HHD_2, were conservatively labeled as normal (Fig. 4e, middle). This likely reflects the limitations of traditional CNV inference in high-purity HHD samples, where global shifts can confound clustering-based baseline identification. Notably, scTransMIL assigned high tumor scores to the B-cell clusters in both HHD samples, consistently identifying malignant patterns that align with the patient-level diagnosis despite the normal status assigned by the computational benchmark (Fig. 4e, right).

Building upon its high-resolution cell-level performance, we evaluated whether scTransMIL's attention mechanism could accurately prioritize key molecular markers of malignancy. By analyzing the attention-score ranking for tumor B cells, we found that scTransMIL not only uncovered the established ALL markers[61] like *CD79A* and *PIM1*[62] within the top 100 genes but also identified *BLK*[63], a known tumor B-cell marker that conventional methods (such as the Wilcoxon Rank-Sum test) miss (Fig. 4f). Our systematic comparison between scTransMIL's attention-based inference and traditional differential expression analysis (DEA) methods reveals several key advantages. We selected normal B cells from three normal samples[58]—PBMMC_1 (GSM3872442), PBMMC_2 (GSM3872443), and PBMMC_3 (GSM3872444)—to serve as control cells, and performed DEA against the B cells in the tumor samples. When analyzing B cells from a single ALL patient (ETV6_RUNX1_4), conventional DEA using the Wilcoxon test identified only *CD79A* and *VPREB3* due to inherent sample size limitation and sample selection bias, while multi-patient DEA using the Wilcoxon test (incorporating B cells from all ALL patients in the pan-cancer dataset) still failed to detect *BLK*—a well-established tumor B cell marker—despite increased statistical power (Fig. 4g and Supplementary Fig. 7a, b). scTransMIL recovered *BLK* through its full transcriptome attention mechanism, even when analyzing individual samples. This capability stems from scTransMIL's capacity to recognize biologically significant genes through their contextualized genomic patterns rather than relying solely on differential expression, as confirmed by *BLK*'s constant expression levels between tumor and normal samples (Supplementary Fig. 7c, d). The model's ability to identify such functionally important yet non-differential genes indicates an advantage over conventional approaches for comprehensive biomarker discovery.

To biologically validate the functional relevance of the identified gene sets by scTransMIL, we performed comprehensive pathway enrichment analysis, which revealed that the top 20 enriched pathways (FDR < 0.001) were strongly associated with ALL pathogenesis (Fig. 4h and Supplementary Data 2), including immune response regulating signaling pathway, cytokine-mediated signaling pathway, and antigen receptor-mediated signaling pathway. The robust enrichment of these biologically coherent pathways provides validation that scTransMIL's attention mechanism captures functionally relevant gene sets beyond what conventional differential expression analysis can identify, confirming its ability to extract meaningful biological insights from single-cell transcriptomic data.

Deciphering novel AML heterogeneity and subtypes with scTransMIL

To evaluate scTransMIL's generalization capability on diseases beyond its original 30-class cancer types, we performed systematic fine-tuning experiments on an acute myeloid leukemia (AML) dataset, a disease category absent from the pan-cancer training data. While scTransMIL is pre-trained to recognize 30 cancer types, its ability to adapt to previously unseen diseases is essential for expanding its applicability. The AML dataset comprised 35 tumor and 6 normal samples

https://doi.org/10.1038/s41467-026-77538-5

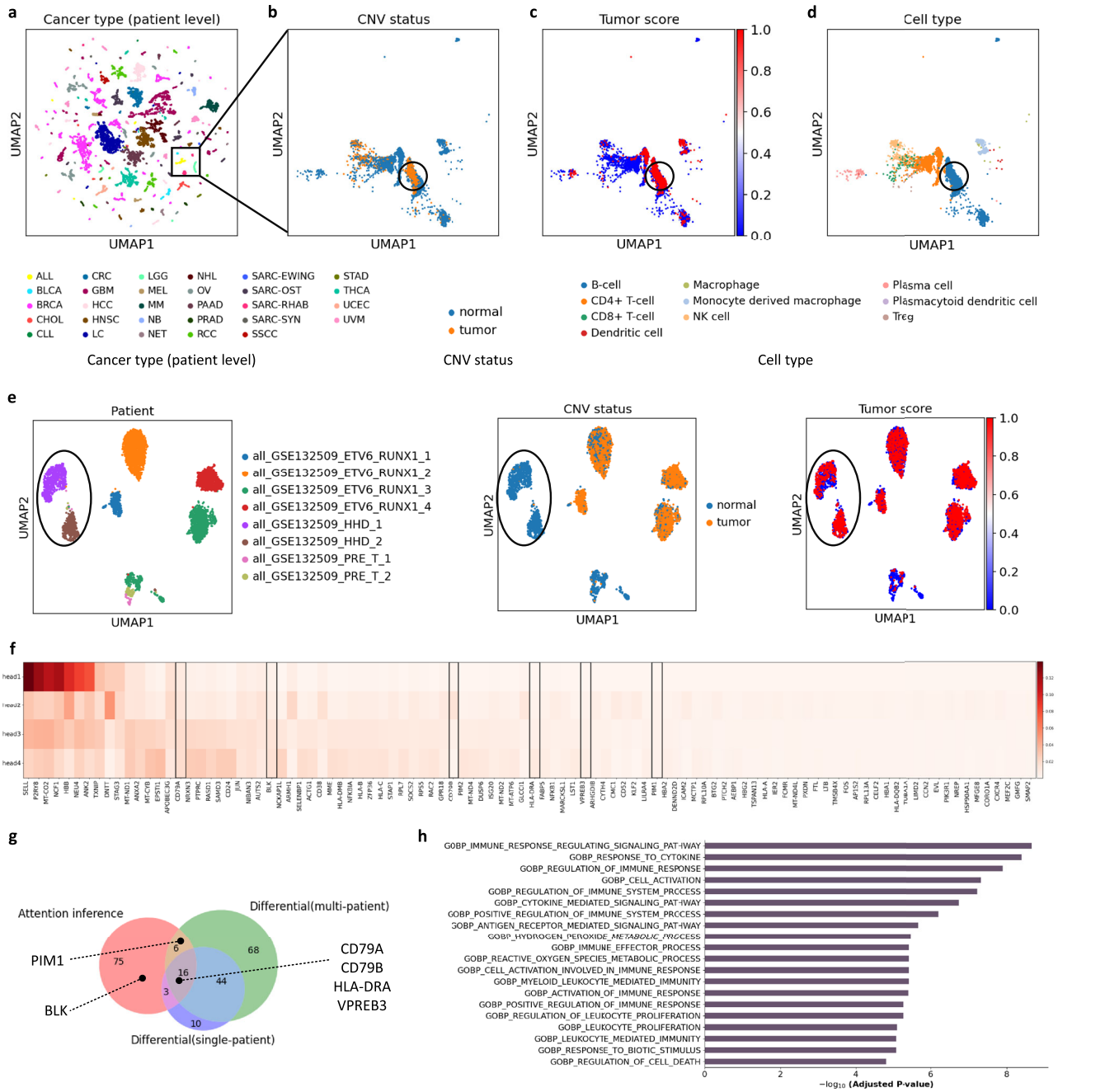


Fig. 4 | Gene-level biological insights inferred by scTransMIL in ALL patient's samples. a UMAP visualization of bag embeddings for the pan-cancer and BRCA datasets colored by cancer types. The black square area represents a selected batch of acute lymphoblastic leukemia (ALL) samples. UMAP visualization of single-cell embeddings for selected ALL patient's samples, colored by CNV-based ground truth annotations distinguishing tumor and normal cells (**b**), tumor scores inferred by scTransMIL (**c**), and cell type annotations (**d**), highlighting a B cell cluster enriched for malignant cells (black circled regions in **b**–**d**). **e** Identification of malignancy in low-CNV signal samples. In B cells from ALL patient's samples, two high hyper diploid (HHD) samples are marked with black ellipses (left). While CNV annotations label them as normal (middle), scTransMIL tumor scores successfully identify the tumor B cells in both samples (right). **f** Heatmap of scTransMIL gene attention scores (top 100) for tumor B cells across four attention heads. Known tumor B cell marker genes are highlighted by black rectangles. **g** Venn diagram comparing genes identified by scTransMIL attention inference with those from differential expression analyses (single-patient and multi-patient settings). scTransMIL identifies *BLK*, a key tumor B cell marker missed by conventional methods. **h** Functional enrichment analysis of the top 100 genes inferred by scTransMIL for tumor B cells, with the top 20 pathways visualized. Functional enrichment *p* values were computed using GSEApy with Fisher's exact test (one-sided), and *p* values were adjusted for multiple comparisons using the Benjamini–Hochberg (BH) method. Source data are provided as a Source Data file.

(Supplementary Fig. 8a, b), with gold-standard cell-level malignant annotations generated through an improved seq-well measurement technique in the original study, which is of high-cost and inefficient[64]. In contrast to these costly single-cell labeling methods, scTransMIL demonstrated data efficiency by achieving robust performance using only a few bag-level tumor/normal labels (20 tumor and 3 normal samples from 21 donors) for fine-tuning and the remaining 15 tumor and 3 normal samples for evaluating. We implemented a strategic binary classification framework during fine-tuning to prioritize fundamental tumor-normal discrimination while preserving the model's capacity to uncover novel biological heterogeneity[65]. To identify the most effective fine-tuning strategy, we compared three experimental

settings: (1) direct application of the pre-trained model without fine-tuning; (2) fine-tuning using only the Bag aggregator while freezing the whole Instance encoder; and (3) fine-tuning the last two transformer blocks and the Bag aggregator, while freezing the remaining layers. As shown in Fig. 5a, we found that selectively fine-tuning the last two transformer blocks alongside the Bag aggregator (while freezing fewer parameters) yielded optimal performance on the previously unseen disease. Furthermore, to assess the label efficiency of scTransMIL, we performed a down-sampling analysis to explore the minimum cohort size required for reliable diagnosis (Supplementary Fig. 8c). We observed that the model's performance (Accuracy and F1 score on the same test set) plateaus with as few as 10 tumor and 3 normal training samples, indicating that our model generalizes well even under limited supervision.

To interpret the representation ability of scTransMIL, we extracted the instance embeddings of all cells from the AML dataset and compared with the original single-cell expression profiles. Notably, the fine-tuned scTransMIL model generated embeddings that more clearly recapitulated a continuous hematopoietic differentiation trajectory from hematopoietic stem cells (HSC) to monocytes (Mono) in normal bone marrow samples (Fig. 5b). This pattern is consistent with and refines the transcriptomic gradient observed in the original single-cell expression profiles (Supplementary Fig. 8b), demonstrating that scTransMIL effectively preserves fundamental biological relationships while providing a more structured low-dimensional representation for cellular transition analysis. Furthermore, tumor scores inferred by the post-fine-tuned scTransMIL model showed markedly improved alignment with the ground truth malignancy annotations compared to the pre-fined-tuned model (Fig. 5c), demonstrating scTransMIL's ability to accurately differentiate tumor and normal cells using only sample-level labels. Expanding beyond binary classification, we identified six distinct AML malignant subtypes (each comprising >1000 tumor cells) (Fig. 5d), by applying a conservative tumor score threshold of >0.2 followed by kNN-based annotation relative to normal differentiation states. These six AML subtypes identified by scTransMIL precisely recapitulated the established AML differentiation trajectory, faithfully capturing the continuum from primitive malignant HSC-like cells through GMP-like cells to terminally differentiated Mono-like cells (Fig. 5e). This recapitulation of known biological trajectories[64] provides validation of scTransMIL's ability to resolve clinically relevant heterogeneity.

Longitudinal analysis of AML patient's samples across treatment timepoints revealed clinically meaningful shifts in tumor cell burden and subtype composition during treatment, exemplified by two representative cases: patient AML707B maintained predominantly GMP-like subtype, consistent with low myeloid differentiation marker expression indicated by flow cytometry in the original study[64], while patient AML419A showed Mono-like subtype, aligning with high myeloid differentiation marker expression (Fig. 5f). These findings collectively demonstrate scTransMIL's capability to simultaneously resolve fundamental biological processes (normal differentiation), disease states (malignant transformation), clinically relevant tumor evolution trajectory and relevant shifts in patient treatment response.

Moreover, to validate the clinical relevance of scTransMIL's AML subtyping, we performed a comprehensive survival analysis on an independent cohort of 671 AML patients[66] containing bulk RNA-seq data only. Subtype-specific gene sets (top 100 genes per subtype) identified by the attention mechanism of scTransMIL (Supplementary Fig. 9) showed partial overlap with those established marker genes in the original single-cell AML study[64]. To stratify this bulk RNA-seq cohort, each patient was uniquely assigned to an AML subtype. Specifically, for each subtype, we selected the top 50 patients who exhibited the highest average expression of the top 30 attention-prioritized signature genes, strictly excluding any patient overlap. This filtering process yielded six distinct subtype patient sets (totaling 300 patients). Subsequent hierarchical clustering performed on the centroids of these six groups (Supplementary Fig. 10, right dendrogram) macroscopically recapitulated the established AML differentiation hierarchy[64]: differentiated myeloid states (cDC-like, Mono-like, and ProMono-like) were distinctly separated from the more primitive or progenitor-like states (GMP-like, HSC-like, and Prog-like).

Kaplan-Meier survival analysis was conducted on each subtype of AML patients, and showed that patients classified under the HSC-like subtype exhibited significantly worse prognosis compared to those in the GMP-like subtype (Supplementary Fig. 11). This finding is consistent with known research[64] that malignant HSCs are more resistant to chemotherapy, evade immune clearance, and contribute to relapse[67]. Functional enrichment analysis of subtype-specific genes provided mechanistic explanations for these clinical observations: HSC-like AML-specific genes were strongly associated with immune evasion pathways and correlated with poor survival, e.g., "GSE40666 UNTREATED VS IFNA STIM STAT1 KO CD8 TCELL 90MIN DN" (Supplementary Data 3 and Supplementary Fig. 12), while GMP-like AML-specific genes were associated with differentiation pathways and correlated with better treatment response. These results demonstrate the potential of scTransMIL in subtype-specific gene mining.

Discussion

The high resolution of single-cell sequencing data can bring about significant changes to the clinical diagnosis of cancer and the research on its mechanisms. Currently, patient-level phenotypic labels are the available meta-information in the samples. However, methods based on bulk data cannot detect malignant cells at the single-cell resolution, making it difficult to apply them in the screening of cancer and the inference of its mechanisms. At the same time, accurate phenotypic annotation at the single-cell level is still lacking, which also hinders the application of existing extensive single-cell data-based methods in the screening of cancer and the analysis of its heterogeneity. In order to connect patient-level disease states with single-cell transcriptome data, we proposed scTransMIL for single-cell data based on multi-instance learning.

The scTransMIL proposed in this study is a deep learning model based on single-cell transformer and multi-instance learning, which can effectively connect the cancer states of patient's samples with single-cell atlases. scTransMIL achieves 3 functions simultaneously: (1) Given a sample from a donor, accurately predict the cancer phenotype of this sample. (2) Infer the disease-relatedness of each cell in the sample at the single-cell resolution to study the cellular processes driving the disease phenotype. (3) Infer genes specific to tumor cells across the full transcriptome to promote research on potential biomarkers.

Extensive experiments proved the versatile applicability and strong performance of our method. scTransMIL achieved competitive performance in benchmark tests. scTransMIL can accurately predict the cancer types for unseen samples and performs well even in predicting metastatic samples, which is of significance for the clinical diagnosis of cancers of unknown primary origin. In addition, the disease-relatedness inferred by scTransMIL at the single-cell resolution is consistent with the ground truth of malignant cells. Because the tumor score is derived from transcriptomic patterns associated with the bag-level tumor label, non-malignant cells in tumor-associated or other perturbed microenvironments may also receive elevated scores.

In terms of biological insights, scTransMIL has significant advantages. Traditional gene-level analysis methods rely on differential analysis and are easily affected by sample selection, while scTransMIL takes into account the full transcriptome space, and its attention mechanism can provide global gene-level biological insights. In the analysis of samples from patients with acute lymphoblastic leukemia, scTransMIL successfully recommended markers for tumor B cells. For the analysis of new cancer types, scTransMIL shows good adaptability

https://doi.org/10.1038/s41467-026-77538-5

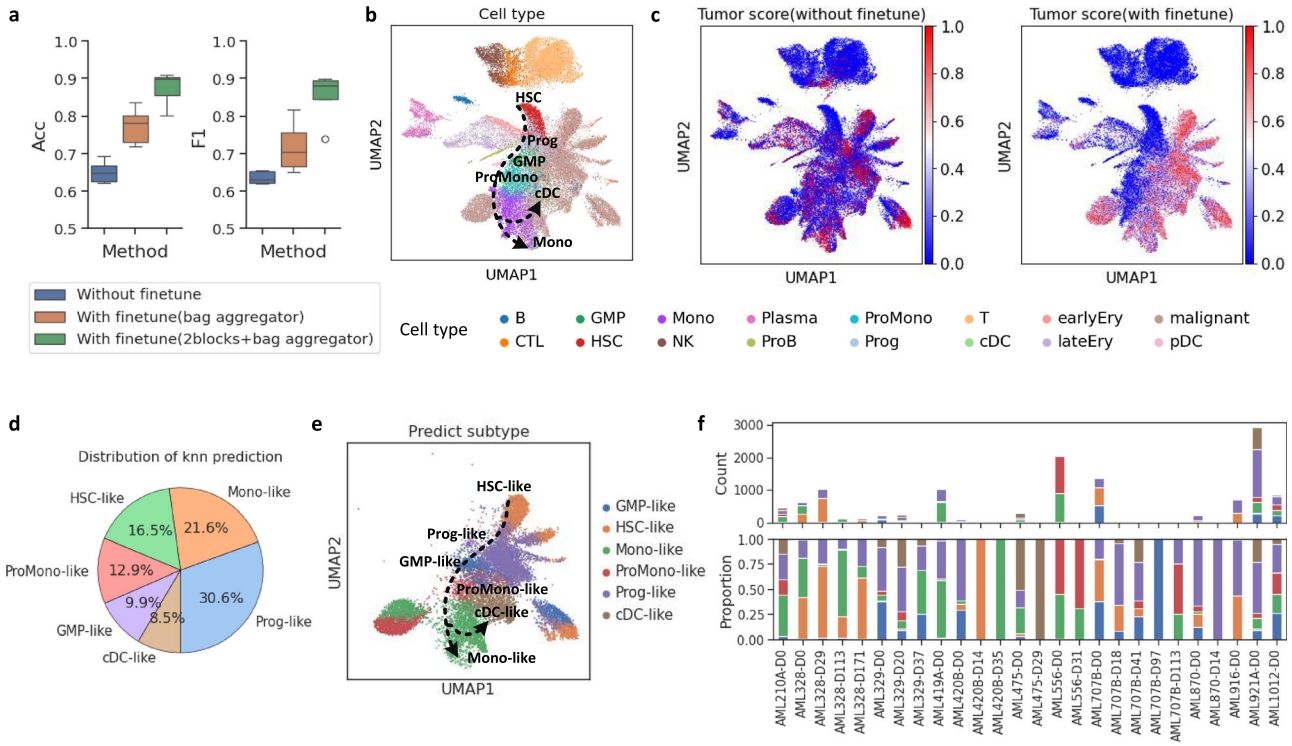


Fig. 5 | scTransMIL enables novel cancer heterogeneity analysis on AML. **a** Performance comparison of scTransMIL under three settings on the acute myeloid leukemia (AML) dataset: without fine-tuning, fine-tuning with bag aggregator only, and fine-tuning with partial instance encoder (the last two transformer blocks) plus bag aggregator. $n = 7$ independent random data splits. In the box plots, the center line denotes the median, the box bounds denote the 25th and 75th percentiles (interquartile range), and the whiskers extend to the minimum and maximum values. **b** UMAP visualization of single-cell instance embeddings generated by scTransMIL for the AML dataset, colored by ground-truth normal cell type annotations. Black dotted lines and arrows indicate the continuous hematopoietic differentiation trajectories. **c** UMAP visualization of single-cell instance embeddings, colored by tumor scores predicted by scTransMIL before (left) and after (right) fine-tuning. **d** Proportional distribution of malignant cells within 6 predicted AML subtypes. Tumor cells were assigned to subtypes resembling normal differentiation states (e.g., HSC-like subtype) using kNN-based classification on instance embeddings. **e** UMAP visualization of malignant cells only, colored by the inferred AML subtypes, revealing a trajectory consistent with hematopoietic lineage progression. **f** Total tumor cell number (top) and AML subtype composition (bottom) across 35 AML samples from 16 patients at different treatment stages. Source data are provided as a Source Data file.

and interpretability. Taking acute myeloid leukemia as an example, minimal fine-tuning scTransMIL using only a few patients' sample-level labels enabled: (1) reconstruction of differentiation trajectories, (2) identification of clinically distinct AML subtypes at the single-cell resolution.

Despite its performance, several limitations of scTransMIL must be acknowledged. First, our primary validation utilized CNV-based malignancy annotations as a benchmark. While the model showed promise in AML, its ability to detect malignant cells in tumors that lack both genomic instability and distinct transcriptomic "context" remains limited. Second, scTransMIL performance on very rare cancers or cohorts with severe class imbalance has not been fully established. Third, there are inherent risks of false predictions. A false negative may lead to a delayed diagnosis of a life-threatening malignancy, while a false positive could trigger unnecessary invasive follow-up procedures.

Methods

In this section, we first introduce datasets, preprocessing, and the model architecture of scTransMIL. Then, we present the usage of scTransMIL, including training, inference, and fine-tuning. Next, we introduce the comparative methods involved in the experiments. Finally, we provide the detailed information of the complete application setup. scTransMIL first classifies patient's samples into tumor bags or normal bags (the first stage) and further distinguishes the cancer type of tumor bags (the second stage). In the first stage, the instance encoder generates the initial embedding for each instance, the gated encoder outputs the gated embedding and tumor score for each instance, and the bag aggregator integrates the gated embeddings to obtain the bag embedding for binary classification. For tumor bags, the second stage employs a multi-classifier to predict the cancer type. During training, the model is optimized using both bag loss and adaptive clustering loss.

Ethics statement

This study used only publicly available, de-identified datasets obtained from published studies and public data repositories; no new data from human participants or animals were collected. All original studies had obtained the relevant ethical approvals and informed consent as described in their respective publications. Accordingly, no additional ethical approval was required for this secondary computational analysis, and our research complies with all relevant ethical regulations.

Data acquisition

Pan-cancer dataset. In the pan-cancer dataset[49], 1563 samples were collected, including single-cell transcriptome sequencing data of 4 million cells. The feature space covers 30,000 genes. The dataset is composed of 104 publicly available datasets and encompasses 30 different cancer types. Detailed metadata for each dataset, including source, cancer type, donor count, and cellular composition, is provided in Supplementary Data 4. Each sample is annotated with a tumor or normal label. For tumor samples, the dataset further annotates the cancer type and ensures that the annotated cancer type is paired with the tissue of cancer origin. At the single-cell resolution, the pan-cancer

dataset also annotates each cell with the corrected copy number variation inference results.

BRCA dataset. We collected an external validation dataset[53] to further evaluate the generalization ability of the model. The BRCA dataset contains 400 thousand cells from 37 samples corresponding to 37 breast cancer (BRCA) patients. In this dataset, all samples are malignant breast cancer, so all bags are assigned the BRCA label. At the cellular resolution, the dataset has annotated malignant cells and other cell types.

AML dataset. To demonstrate the analysis of new cancer type other than the 30 cancer types in the pan-cancer dataset, we took acute myeloid leukemia (AML) as an example and selected the AML dataset[64]. This dataset contains 35 tumor bags and 6 normal bags, with a total of 30,000 single cells. At the single-cell resolution, the dataset includes annotation information of malignant cells and other cell types, and at the patient's sample resolution, it contains the treatment time.

Bulk AML dataset. To validate the AML subtype-specific genes obtained from the AML dataset, we used an independent bulk AML RNA-seq dataset[66] for survival analysis. This dataset measured 671 AML patients and recorded the time when the death events occurred.

Data preprocessing

For all datasets included in this study, we utilized gene expression count matrices acquired from public repositories rather than processing raw sequencing data. We implemented a preprocessing pipeline based on the Modal-NexT framework[68]. Specifically, the count matrices were updated for each single-cell dataset, and genes within each individual cell were converted into a rank-based representation to mitigate technical variation and ensure cross-dataset compatibility.

To overcome the computational bottlenecks, we introduced a sub-sample strategy during data preparation. A scRNA-seq sample contains tens of thousands of cells. Processing an entire sample simultaneously in a single forward pass would far exceed GPU memory capacities. Therefore, we computationally partitioned each original sample into multiple sub-samples, with a maximum size of 256 cells per sub-sample. During the training phase, these sub-samples were assigned labels based on the cell-level CNV status. Following the MIL assumption, a sub-sample was labeled as a "tumor sub-sample" if it contained at least one malignant cell, and as a "normal sub-sample" only if all its constituent cells were normal. This partitioning strategy was applied consistently across all datasets. All comparison methods were also trained under this setting.

We acknowledge that CNV-based inference is not flawless, which inevitably introduces label noise into the sub-samples. Nevertheless, our experimental results demonstrate that the scTransMIL architecture is robust to such noisy supervision. Crucially, during the inference (testing) phase, the evaluation is strictly aggregated back to the original sample level. For a given test sample, the model processes all its constituent sub-samples independently. If at least one sub-sample is predicted as a tumor, the entire sample is classified as tumor-positive. For multi-class classification, voting is used to determine the entire sample's cancer type. The final performance metrics are then calculated by comparing these aggregated predictions against the true diagnoses (original sample's diagnostic result can serve as ground truth label). The test performance on unseen samples confirms that our model effectively learns robust patient-level classification capabilities from the sub-sample training strategy. For the consistent description, we use the bag to represent samples or sub-samples.

Instance encoder

Given a donor's sub-sample (bag), which contains multiple single cells, the instance encoder generates an initial embedding for each single cell. For a single cell, we use rank value encoding to convert the gene expression levels of the cell into a gene sequence. The gene sequence can be regarded as a sentence, with genes acting as words. Rank value encoding takes into account the full transcriptome characteristics of the single cell. We filter out genes with an expression level of 0, thus adaptively reducing the length of the gene sequence for each single cell. Then, we rank the remaining genes according to their gene expression levels to obtain a gene sequence (genes, as words, are mapped to IDs in the vocabulary). Since the gene sequence does not involve specific expression levels, batch effects can be alleviated. The instance encoder consists of 6 transformer encoder units, and each unit is composed of a self-attention layer and a feed-forward neural network layer[39]. The instance encoder transforms genes into a 256-dimensional space through the transformer. This space encodes the characteristics of the gene in a specific cellular environment (gene sequence) through self-attention. The initial embedding of any cell is generated by averaging the embeddings of each gene in the gene sequence, resulting in a 256-dimensional initial cell embedding. For scTransMIL(PCA) and scTransMIL(FM), we use the PCA cell embedding and the cell embedding output by the foundation model respectively to replace the initial cell embedding output by the instance encoder.

Gated encoder

In a bag, based on the initial cell embeddings, the gated encoder updates the cell embeddings through a gating mechanism, and outputs the gated embeddings and tumor scores. The gating mechanism can filter out the noise signals in the initial cell embeddings that are irrelevant to the category of the bag[69]. Let the initial cell embedding of cell (instance) i be $h_{init}^{(i)} \in \mathbb{R}^{256}$, and the corresponding gated embedding and tumor score be $h_{gate}^{(i)} \in \mathbb{R}^{256}$ and $a^{(i)} \in \mathbb{R}$ respectively.

The initial cell embedding of instance i is input into the gated encoder. First, it passes through a non-linear transformation module composed of a linear transformation layer and a tanh activation function to obtain the updated cell embedding $h_1^{(i)} \in \mathbb{R}^{256}$. Then, it goes through another non-linear transformation module consisting of a linear transformation layer and a sigmoid activation function to output the filtering vector $h_2^{(i)} \in \mathbb{R}^{256}$. The filtering vector acts on the updated cell embedding and outputs the gated embedding, which is used to retain informative features and suppress non-informative features:

$$h_{gate}^{(i)} = h_1^{(i)} \odot h_2^{(i)}, \tag{1}$$

where $\odot$ represents element-wise multiplication. Based on the binary classification scenario in the first stage, each bag will be assigned either a tumor label (1) or a normal label (0). We use the tumor score to measure the correlation between cell i and the tumor label. We input the gated embedding into $linear(\cdot)$ and output the tumor score:

$$a^{(i)} = linear(h_{gate}^{(i)}). \tag{2}$$

Bag aggregator

The theoretical foundation of scTransMIL follows the standard MIL assumption: a patient-level "bag" is considered positive if it contains at least one malignant instance. To implement this logic in a computationally efficient and differentiable manner, we utilize an attention-based weighted sum rather than a rigid Max-pooling operator. In this architecture, the inferred tumor scores act as adaptive attention weights that prioritize cells driving the global diagnostic signal. By learning to suppress non-malignant background features and amplify tumor-associated signatures, the model can effectively approximate the "OR" logic of classic MIL while enabling stable gradient flow and the capture of intra-tumoral heterogeneity.

To be specific, for a bag j, assume that it contains m instances (the number of instances in different bags is variable). We use the instance encoder and the gated encoder to obtain the gated embeddings $\{h_{gate}^{(i)}\}_{i=1}^{m}$ and tumor scores $\{a^{(i)}\}_{i=1}^{m}$ of all instances in the bag. The bag aggregator is used to integrate the instance representations in the current bag j and output a bag embedding $h_{bag}^{(j)} \in \mathbb{R}^{256}$. We achieve the integration by weighting the gated embeddings with the tumor scores:

$$h_{bag}^{(j)} = \sum_{i=1}^{m} a^{(i)} h_{gate}^{(i)}. \tag{3}$$

In the first stage, the bag embedding is input into a linear transformation and a binary softmax activation function to obtain the binary classification result $y_{1st}^{(j)}$. The bag will be assigned either a tumor label or a normal label. If the bag is assigned the tumor label, the bag embedding is input into another linear transformation and a multi-class softmax activation function to obtain the multi-classification result $y_{2nd}^{(j)}$, and the bag will be assigned a cancer type.

Loss

The model is trained under the assumption of multiple-instance learning (MIL), which is a weakly supervised learning process. We only need to provide the bag labels. With the goal of bag classification, the model can infer the correlation between instances and bag labels by updating its parameters. For bag-level supervision, let the true label of bag j be $y_{bag}^{(j)}$. Taking the binary classification of tumor or normal as an example, we define the bag-level loss $y_{\text{bag}}^{(j)}$ using the binary cross-entropy loss:

$$L_{bag}^{(j)} = -[y_{bag}^{(j)} log(y_{1st}^{(j)}) + (1 - y_{bag}^{(j)}) log(1 - y_{1st}^{(j)})]. \tag{4}$$

For the cancer type identification in the second stage, we use the multi-class cross-entropy loss:

$$L_{bag}^{(j)} = -y_{bag}^{(j)} log(y_{1st}^{(j)}). \tag{5}$$

To prevent the model from overfitting on a few easily classifiable positive instances, the adaptive clustering loss is used to assist the model in distinguishing between positive and negative instances. To balance the learning of positive and negative embeddings, we construct the adaptive clustering loss based on the SVM loss. Since the true instance-level labels do not exist, we assign pseudo-labels according to the tumor scores. A high tumor score indicates a high probability that the instance is positive, while a low tumor score indicates a high probability that the instance is negative[43].

For bag j, its true binary classification label is $y_{bag}^{(j)} \in \{0, 1\}$, where 1 represents tumor and 0 represents normal. All the tumor scores in this bag are sorted. We select the B instances with the lowest tumor scores and assign them negative labels, and select the B instances with the highest tumor scores and assign them positive labels. In this study, we set $B = 8$ as a fixed hyperparameter. This fixed selection ensures that the loss specifically constrains the most discriminative and high-confidence cellular embeddings, preventing the introduction of noise from ambiguous intermediate instances.

For an instance i in bag j, we add a linear transformation $W_{\text{inst}} \in \mathbb{R}^{2\times 256}$ to define the instance's clustering assignment probability as $P^{(i)}$:

$$P^{(i)} = W_{inst} h_{gate}^{(i)}. \tag{6}$$

We adopt the smoothed SVM loss. Given that the predicted probability of instance i in bag j is $P^{(i)} = [p_0^{(i)}, p_1^{(i)}]$ and the assigned pseudo-label is $y_{inst}^{(i)} \in \{0, 1\}$, the SVM loss is:

$$L(P^{(i)}, y_{inst}^{(i)}) = \tau log[\sum_{k=0}^{1} \exp(\frac{1}{\tau}(\alpha \cdot \mathbb{I}(k \neq y_{inst}^{(i)}) + p_k^{(i)} - p_{y_{inst}^{(i)}}^{(i)}))] \tag{7}$$

where τ is the temperature coefficient, with a default value of 1.0, which is used to control the smoothness of the SVM loss function to facilitate gradient updates. α is the specified margin, with a default value of 1.0, which is used to control the interval between the predicted scores of different classes. $\mathbb{I}$ is the indicator function, which takes the value of 1 when $k \neq y_{inst}^{(i)}$, and 0 otherwise.

Since instances with high tumor scores do not necessarily all correspond to malignant cells, and instances with low tumor scores do not necessarily all correspond to benign cells, there must be noise in the labels created for supervising the instance clustering task during the training process. However, the SVM loss can reduce overfitting when the data labels are noisy.

For any bag j, we only need to select $2B$ instances to calculate the adaptive clustering loss $L_{\text{ada}}^{(j)}$:

$$L_{ada}^{(j)} = \frac{1}{2B} \sum_{i=1}^{2B} L(P^{(i)}, y_{inst}^{(i)}). \tag{8}$$

Therefore, the loss on bag j is:

$$L^{(j)} = \beta L_{bag}^{(j)} + (1 - \beta) L_{ada}^{(j)}, \tag{9}$$

where β is the coefficient of the loss function, with a default value of 0.7.

Training

We trained scTransMIL on a pan-cancer dataset composed of 104 publicly available cancer datasets[49]. The training resources covered 1563 samples, 4 million single cells, and 30,000 genes. To make a bag available within the forward computation pipeline, we obtained 20,000 sub-samples following "Methods" subsection "Data preprocessing" (Supplementary Fig. 3a). Only the sub-sample level had cancer labels, which included binary classification scenario (tumor or normal) and multi-classification scenario (cancer type). We regarded sub-samples as bags. Each donor usually had multiple sub-samples. Therefore, there were two ways to assign labels to bags. For the first label assignment method (Sample setting), bag labels were low noisy (bag labels are derived from CNV status at single-cell resolution, refer to "Methods" subsection "Data preprocessing"). For the second label assignment method (Donor setting), we assigned labels at the donor level. If a donor was a cancer patient, all bags belonging to this donor were assigned the tumor label. This method would introduce noise and could be used to test the model's learning ability in extreme situations, such as whether the model focused on causal features instead of statistically correlated features. scTransMIL was trained and inferred in the first label assignment method by default.

To ensure the robustness and reproducibility of our findings, all model evaluations were performed using a repeated random split validation protocol ($n = 7$). For each iteration, the dataset was randomly divided into a training set (80% of the donors), a validation set (10% of the donors), and a test set (10% of the donors). This ensures that the donors in the test set or validation set do not overlap with those in the training set, thus directly evaluating the generalization ability of the model on unseen donors.

During training, bags were randomly sampled with a batch size of 1. The multi-nomial sampling probability of each bag was inversely proportional to the frequency of its true category to mitigate the class imbalance problem in the training set. The model parameters were randomly initialized and trained end-to-end using bag-level labels. The model parameters were updated by the Adam optimizer with a

learning rate of 2×10^{-4} and a weight decay of 1×10^{-5}. The model was trained for at least 30 epochs. If the early stopping criterion was not met, it was trained for a maximum of 200 epochs. We monitored the validation loss during training. If the validation loss did not decrease for 10 consecutive epochs, we used early stopping. We selected the model with the lowest validation loss for testing on the test set.

Inference

Given a sub-sample (bag), scTransMIL can predict whether the bag is a tumor sub-sample or a normal sub-sample and infer the tumor score of all instances in the bag. If a bag is predicted to be a tumor sub-sample, scTransMIL further predicts the cancer type of the bag. For a given test sample, the model processes all its constituent sub-samples independently. If at least one sub-sample is predicted as a tumor, the entire sample is classified as tumor-positive. For multi-class classification, voting is used to determine the entire sample's cancer type. We first tested the binary classification of scTransMIL on pan-cancer dataset (internal dataset), and then selected all samples with true tumor labels to test the performance of scTransMIL in predicting cancer types in the second stage. To further test the generalization ability of the model, we prepared an external validation BRCA dataset as the test set, which contains 37 samples (1500 sub-samples) from breast cancer patients[53].

In addition, we utilized the attention mechanism of scTransMIL to infer genes specific to the target cell group. scTransMIL returns the attention of the last transformer layer corresponding to each instance from the instance encoder, and we calculate the mean of the attention in all heads. After collecting all target instances, we averaged the mean attention of all target instances as the attention specific to the target cell group. According to the attention values, we ranked the genes and selected the top 100 genes as the genes specific to the target cell group.

Fine-tuning

Given that the pan-cancer training set encompasses 30 types of cancers, for unseen cancer type such as acute myeloid leukemia (AML), we can perform analysis of new disease by fine-tuning scTransMIL. Taking the AML dataset as an example, this dataset includes 5 healthy donors and 16 AML patients[64]. The annotation information used for fine-tuning only comes from the cancer states (tumor or normal) of these 21 donors. To evaluate the performance of scTransMIL in scenarios with limited donor labels, we employed Donor setting to assign labels to sub-samples, which were then utilized by scTransMIL. We let scTransMIL only perform fine-tuning at the binary classification stage to avoid overfitting. We froze the parameters of the instance encoder, set the maximum number of training epochs to 30, and kept other settings consistent with those in the pan-cancer training.

Comparison methods

To evaluate the performance of scTransMIL, we compared it against four established methods representing distinct computational paradigms for linking patient-level disease states with single-cell transcriptomics. These methods include pseudo-bulk aggregation, joint latent space learning, and both instance-based and bag-based MIL. A comprehensive summary of the core algorithms, software versions, implementation sources, and key hyperparameters is provided in Supplementary Data 5.

KIDA (pseudo-bulk aggregation). KIDA[36] represents methods that simplify single-cell data into a pseudo-bulk format. For each patient's sample, single-cell expressions are averaged to generate a bulk profile. KIDA then initializes its network parameters based on biological pathway annotations to transform gene expression into functional patterns. A self-attention mechanism is employed to learn the interactions between these patterns, ultimately outputting sample-level predictions. We utilized the original Python implementation provided by the authors.

DEGAS (joint latent space learning). DEGAS[37] focuses on the simultaneous integration of patient's samples and single-cell data. It utilizes a Maximum Mean Discrepancy (MMD) loss function to minimize the distribution gap between entities at these two different resolutions. By learning a joint latent space, the framework allows for the transfer of patient labels to single-cell entities. The computational complexity of DEGAS is primarily driven by this multi-resolution data integration. We used the original R package provided by the authors.

scIDST (instance-based MIL). scIDST[40] is an instance-based MIL framework that assigns the bag-level (patient) label to all constituent cells as a pseudo-label. It utilizes a fully connected network to learn the mapping between individual instances and these pseudo-labels. To mitigate the technical noise and biological heterogeneity inherent in instance-level labels, scIDST introduces a probabilistic labeling strategy. We utilized the original Python implementation from the official repository.

scVI+MIL (bag-based MIL). This baseline serves as a representative implementation of bag-based MIL. The process involves two stages: first, we utilized scVI[70] for the unsupervised learning of instance-level representations to capture the underlying latent structure of the single-cell data. Second, a standard bag-based MIL process was employed to map these instance representations to bag-level labels. For the MIL component, we adopted the attention-pooling mechanism implemented in MultiMIL[41], which we adapted for single-omics data using the official scvi-tools and MultiMIL codebases.

For all methods, we adhered to the original authors' recommended default settings to maintain the integrity of the comparison. Furthermore, we conducted systematic fine-tuning on our internal validation sets to ensure that each baseline reached its optimal performance for the specific pan-cancer and clinical datasets used in this study.

Implementation of scTransMIL variants

To evaluate the specific contributions of the rank-based Transformer encoder, we developed two architectural variants for comparison:

scTransMIL(PCA). In this variant, cell-level embeddings are generated using linear dimensionality reduction rather than a Transformer. We performed principal component analysis (PCA) on the log-normalized expression matrix of the highly variable genes, selecting the top [e.g., 512] principal components to match the embedding dimension of the standard scTransMIL. These PCA-derived embeddings were then passed directly to the gated encoder and attention-pooling modules.

scTransMIL(FM). This variant assesses the performance of a value-based foundation model (FM) within our MIL framework. We employed the scGPT architecture and loaded the official "whole-human" pre-trained weights, which were derived from a massive atlas of 33 million human cells. Unlike the rank-based encoding in scTransMIL, scTransMIL(FM) processes absolute expression values using a value-based Transformer. The 512-dimensional embeddings extracted from the last layer of scGPT were utilized for patient-level diagnostic prediction.

Application

Identification of target cell cluster-specific genes. In the single-cell resolution atlas, for any given instance, the instance encoder of scTransMIL returns the attention of the last transformer layer. We calculate the average for all heads to obtain the gene attention score specific to the current instance. Further, we collect all the instances in

the target cell cluster and calculate the average attention of these instances. We rank the genes according to the attention scores and select the top 100 genes as the genes specific to the target cell cluster. In this study, we set two groups of target cell clusters respectively: the tumor cell cluster of the specified cell type and the normal cell cluster of the specified cell type.

Differential expression analysis. In this study, we selected all the samples from patients in the pan-cancer dataset and conducted differential analysis by comparing the normal samples. The differential analysis followed the standard procedure of Wilcoxon Rank-Sum test.

Functional enrichment analysis. Functional enrichment analysis of the top 100 genes prioritized by scTransMIL was performed using GSEApy (gseapy.enrichr). Enrichment significance was assessed with a one-sided Fisher's exact test, and the resulting p values were adjusted for multiple comparisons using the Benjamini–Hochberg (BH) method; the top 20 enriched pathways were visualized.

Cancer subtype-specific genes analysis. We used an independent bulk dataset to validate the cancer subtype-specific genes inferred by the attention mechanism of scTransMIL. First, we defined the genes specific to each cancer subtype based on attention results, with 100 specific genes (gene set) for each subtype. In the bulk data, we selected the samples with a relatively high average expression level of a certain gene set as the samples related to the current cancer subtype. Furthermore, we conducted a Kaplan–Meier survival analysis based on the stratification results of the samples in the bulk data and the recorded occurrence time of death events for each sample.

Visualization. To evaluate the latent representations learned by scTransMIL, single-cell visualizations (UMAP) were generated directly from the model's instance embeddings without the application of external batch correction or standard scaling normalization methods. These embeddings were processed using the Scanpy library, where the neighborhood graph was constructed via scanpy.pp.neighbors and the final UMAP projections were generated using scanpy.tl.umap.

Statistics and reproducibility

No statistical method was used to predetermine sample size. Model performance was evaluated using a repeated random split protocol with $n = 7$ independent random splits of the donor cohort into training (80%), validation (10%), and test (10%) sets, ensuring non-overlapping donors across sets. No data were excluded from the analyses. The assignment of donors to training, validation, and test sets was randomized across the seven independent splits; no other experimental randomization was applicable to this computational study. The investigators were not blinded to allocation, as all analyses were fully automated, label-agnostic during model inference, and evaluated against fixed ground-truth annotations, so blinding was not relevant to the outcome assessment. All reported metrics (Accuracy, AUC, F1 score) are summarized across the $n = 7$ splits.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All datasets analyzed in this study are publicly available and were obtained from public repositories; no data are under restricted access. The single-cell pan-cancer dataset is available at https://cellatlas.kaist.ac.kr/ecosystem/. The thyroid cancer dataset is available in the Gene Expression Omnibus (GEO) under accession code GSE184362. The external breast cancer (BRCA) dataset is available at CELLxGENE (https://cellxgene.cziscience.com/e/34f5307e-7b4d−4a48-b68f-2ba844c6414b.cxg/). The acute myeloid leukemia (AML) single-cell dataset is available in GEO under accession code GSE116256. The bulk AML dataset used for survival analysis is available in dbGaP under accession code phs001657.v2.p1. Source data are provided with this paper.

Code availability

The codes are implemented in https://github.com/TencentAILabHealthcare/scTransMIL. To provide a permanent reference to the exact version of the code used in this study, the GitHub repository has been archived on Zenodo (https://doi.org/10.5281/zenodo.21151067)[71].

References

1. Bunne, C. et al. How to build the virtual cell with artificial intelligence: priorities and opportunities. *Cell* **187**, 7045–7063 (2024).
2. Rood, J. E., Maartens, A., Hupalowska, A., Teichmann, S. A. & Regev, A. Impact of the human cell atlas on medicine. *Nat. Med.* **28**, 2486–2496 (2022).
3. Su, X. et al. Interpretable identification of cancer genes across biological networks via transformer-powered graph representation learning. *Nat. Biomed. Eng.* 1–19 https://doi.org/10.1038/s41551-024-01312-5 (2025).
4. Topol, E. J. Learning the language of life with AI. *Science* **387**, eadv4414 (2025).
5. Heimberg, G. et al. A cell atlas foundation model for scalable search of similar human cells. *Nature* 1–10 https://doi.org/10.1038/s41586-024-08411-y (2024).
6. Szałata, A. et al. Transformers in single-cell omics: a review and new perspectives. *Nat. Methods* **21**, 1430–1443 (2024).
7. Van Dijk, D. et al. Recovering gene interactions from single-cell data using data diffusion. *Cell* **174**, 716–729 (2018).
8. Huang, M. et al. SAVER: gene expression recovery for single-cell RNA sequencing. *Nat. Methods* **15**, 539–542 (2018).
9. Eraslan, G., Simon, L. M., Mircea, M., Mueller, N. S. & Theis, F. J. Single-cell RNA-seq denoising using a deep count autoencoder. *Nat. Commun.* **10**, 390 (2019).
10. Wu, K. E., Yost, K. E., Chang, H. Y. & Zou, J. BABEL enables cross-modality translation between multiomic profiles at single-cell resolution. *Proc. Natl. Acad. Sci. USA* **118**, e2023070118 (2021).
11. Tang, X. et al. Explainable multi-task learning for multi-modality biological data analysis. *Nat. Commun.* **14**, 2546 (2023).
12. Cao, Y. et al. scButterfly: a versatile single-cell cross-modality translation method via dual-aligned variational autoencoders. *Nat. Commun.* **15**, 2973 (2024).
13. Chen, H., Ryu, J., Vinyard, M. E., Lerer, A. & Pinello, L. SIMBA: single-cell embedding along with features. *Nat. Methods* **21**, 1003–1013 (2024).
14. Cao, Z.-J. & Gao, G. Multi-omics single-cell data integration and regulatory inference with graph-linked embedding. *Nat. Biotechnol.* **40**, 1458–1466 (2022).
15. Ashuach, T. et al. MultiVI: deep generative model for the integration of multimodal data. *Nat. Methods* **20**, 1222–1231 (2023).
16. Huizing, G.-J., Deutschmann, I. M., Peyré, G. & Cantini, L. Paired single-cell multi-omics data integration with Mowgli. *Nat. Commun.* **14**, 7711 (2023).
17. Tang, Z., Huang, J., Chen, G. & Chen, C. Y.-C. Comprehensive view embedding learning for single-cell multimodal integration. *Proc. AAAI Conf. Artif. Intell.* **38**, 15292–15300 (2024).
18. Tang, Z. et al. Modal-nexus auto-encoder for multi-modality cellular data integration and imputation. *Nat. Commun.* **15**, 9021 (2024).
19. Chen, Y. & Zou, J. Simple and effective embedding model for single-cell biology built from ChatGPT. *Nat. Biomed. Eng* 1–11 https://doi.org/10.1038/s41551-024-01284-6 (2024).

20. Domínguez Conde, C. et al. Cross-tissue immune cell analysis reveals tissue-specific features in humans. *Science* **376**, eabl5197 (2022).
21. Chen, J. et al. Transformer for one stop interpretable cell type annotation. *Nat. Commun.* **14**, 223 (2023).
22. Stuart, T. et al. Comprehensive integration of single-cell data. *Cell* **177**, 1888–1902 (2019).
23. Yang, F. et al. scBERT as a large-scale pretrained deep language model for cell type annotation of single-cell RNA-seq data. *Nat. Mach. Intell.* **4**, 852–866 (2022).
24. Ergen, C. et al. Consensus prediction of cell type labels in single-cell data with popV. *Nat. Genet.* **56**, 2731–2738 (2024).
25. Cui, H. et al. scGPT: toward building a foundation model for single-cell multi-omics using generative AI. *Nat. Methods* **21**, 1470–1480 (2024).
26. Hao, M. et al. Large-scale foundation model on single-cell transcriptomics. *Nat. Methods* **21**, 1481–1491 (2024).
27. Dohmen, J. et al. Identifying tumor cells at the single-cell level using machine learning. *Genome Biol.* **23**, 123 (2022).
28. Zhu, Q. et al. Single cell multi-omics reveal intra-cell-line heterogeneity across human cancer cell lines. *Nat. Commun.* **14**, 8170 (2023).
29. Sudmant, P. H. et al. Global diversity, population stratification, and selection of human copy-number variation. *Science* **349**, aab3761 (2015).
30. Nofech-Mozes, I., Soave, D., Awadalla, P. & Abelson, S. Pan-cancer classification of single cells in the tumour microenvironment. *Nat. Commun.* **14**, 1615 (2023).
31. Filbin, M. G. et al. Developmental and oncogenic programs in H3K27M gliomas dissected by single-cell RNA-seq. *Science* **360**, 331–335 (2018).
32. Venteicher, A. S. et al. Decoupling genetics, lineages, and microenvironment in IDH-mutant gliomas by single-cell RNA-seq. *Science* **355**, eaai8478 (2017).
33. Huang, C. et al. scCancerExplorer: a comprehensive database for interactively exploring single-cell multi-omics data of human pan-cancer. *Nucleic Acids Res.* **53**, D1526–D1535 (2025).
34. Gavish, A. et al. Hallmarks of transcriptional intratumour heterogeneity across a thousand tumours. *Nature* **618**, 598–606 (2023).
35. Han, Y. et al. TISCH2: expanded datasets and new tools for single-cell transcriptome analyses of the tumor microenvironment. *Nucleic Acids Res.* **51**, D1425–D1431 (2023).
36. Tang, Z. et al. Knowledge-based inductive bias and domain adaptation for cell type annotation. *Commun. Biol.* **7**, 1–10 (2024).
37. Johnson, T. S. et al. Diagnostic Evidence GAuge of Single cells (DEGAS): a flexible deep transfer learning framework for prioritizing cells in relation to disease. *Genome Med.* **14**, 11 (2022).
38. Sun, D. et al. Identifying phenotype-associated subpopulations by integrating bulk and single-cell sequencing data. *Nat. Biotechnol.* **40**, 527–538 (2022).
39. Theodoris, C. V. et al. Transfer learning enables predictions in network biology. *Nature* **618**, 616–624 (2023).
40. Wehbe, F. et al. Inferring disease progression stages in single-cell transcriptomics using a weakly supervised deep learning approach. *Genome Res.* **35**, 135–146 (2025).
41. Litinetskaya, A. et al. Multimodal weakly supervised learning to identify disease-specific changes in single-cell atlases. Preprint at https://doi.org/10.1101/2024.07.29.605625 (2024).
42. Li, H. et al. DT-MIL: deformable transformer for multi-instance learning on histopathological image. In *Medical Image Computing and Computer Assisted Intervention–MICCAI 2021: 24th International Conference, Strasbourg, France, Proceedings, Part VIII 24* 206–216 (Springer, 2021).
43. Lu, M. Y. et al. Data-efficient and weakly supervised computational pathology on whole-slide images. *Nat. Biomed. Eng.* **5**, 555–570 (2021).
44. Shao, Z. et al. Transmil: transformer based correlated multiple instance learning for whole slide image classification. *Adv. Neural Inf. Process. Syst.* **34**, 2136–2147 (2021).
45. Zheng, T., Jiang, K. & Yao, H. Dynamic policy-driven adaptive multi-instance learning for whole slide image classification. In *Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition* 8028–8037 (IEEE, 2024).
46. Xie, Y., Liu, Z., Zhao, J. & Ma, J. P H I M-M I L: multiple instance learning with prototype similarity-guided feature fusion and hard instance mining for whole slide image classification. *Inf. Fusion* **117**, 102847 (2025).
47. Cheng, H. et al. Focus your attention: multiple instance learning with attention modification for whole slide pathological image classification. *IEEE Trans. Circuits. Syst. Video Technol.* **35**, 5791–5804 (2025).
48. Qu, L. et al. Rethinking multiple instance learning for whole slide image classification: a good instance classifier is all you need. *IEEE Trans. Circuits. Syst. Video Technol.* **34**, 9732–9744 (2024).
49. Kang, J. et al. Systematic dissection of tumor-normal single-cell ecosystems across a thousand tumors of 30 cancer types. *Nat. Commun.* **15**, 4067 (2024).
50. Pu, W. et al. Single-cell transcriptomic analysis of the tumor ecosystems underlying initiation and progression of papillary thyroid carcinoma. *Nat. Commun.* **12**, 6058 (2021).
51. Moran, S. et al. Epigenetic profiling to classify cancer of unknown primary: a multicentre, retrospective analysis. *Lancet Oncol.* **17**, 1386–1395 (2016).
52. Tian, F. et al. Prediction of tumor origin in cancers of unknown primary origin with cytology-based deep learning. *Nat. Med.* **30**, 1309–1319 (2024).
53. Klughammer, J. et al. A multi-modal single-cell and spatial expression map of metastatic breast cancer biopsies across clinicopathological features. *Nat. Med.* **30**, 3236–3249 (2024).
54. Agrawal, N. et al. Integrated genomic characterization of papillary thyroid carcinoma. *Cell* **159**, 676–690 (2014).
55. Tirode, F. et al. Genomic landscape of ewing sarcoma defines an aggressive subtype with co-association of *STAG2* and *TP53* mutations. *Cancer Discov.* **4**, 1342–1353 (2014).
56. The Cancer Genome Atlas Network. Comprehensive molecular portraits of human breast tumours. *Nature* **490**, 61–70 (2012).
57. Luecken, M. et al. Benchmarking atlas-level data integration in single-cell genomics. *Nat. Methods* **19**, 41–50 (2022).
58. Caron, M. et al. Single-cell analysis of childhood leukemia reveals a link between developmental states and ribosomal protein expression as a source of intra-individual heterogeneity. *Sci. Rep.* **10**, 8079 (2020).
59. Mantovani, A. et al. The chemokine system in diverse forms of macrophage activation and polarization. *Trends Immunol.* **25**, 677–686 (2004).
60. Mantovani, A., Marchesi, F., Malesci, A., Laghi, L. & Allavena, P. Tumour-associated macrophages as treatment targets in oncology. *Nat. Rev. Clin. Oncol.* **14**, 399–416 (2017).
61. Young, R. M., Phelan, J. D., Wilson, W. H. & Staudt, L. M. Pathogenic B-cell receptor signaling in lymphoid malignancies: new insights to improve treatment. *Immunol. Rev.* **291**, 190–213 (2019).
62. Gökbuget, N. et al. Diagnosis, prognostic factors, and assessment of ALL in adults: 2024 ELN recommendations from a European expert panel. *Blood* **143**, 1891–1902 (2024).
63. Kim, E. et al. Ibrutinib inhibits pre-BCR+ B-cell acute lymphoblastic leukemia progression by targeting BTK and BLK. *Blood* **129**, 1155–1165 (2017).

64. van Galen, P. et al. Single-cell RNA-seq reveals AML hierarchies relevant to disease progression and immunity. *Cell* **176**, 1265–1281 (2019).
65. Tang, Z., Chen, G., Yang, H., Zhong, W. & Chen, C. Y.-C. DSIL-DDI: a domain-invariant substructure interaction learning for generalizable drug–drug interaction prediction. *IEEE Trans. Neural Netw. Learn. Syst.* **35**, 10552–10560 (2024).
66. Bottomly, D. et al. Integrative analysis of drug response and clinical outcome in acute myeloid leukemia. *Cancer Cell* **40**, 850–864 (2022).
67. Saw, P. E., Liu, Q., Wong, P.-P. & Song, E. Cancer stem cell mimicry for immune evasion and therapeutic resistance. *Cell Stem Cell* **31**, 1101–1112 (2024).
68. Tang, Z. et al. Modal-NexT: towards unified heterogeneous cellular data integration. *Inf. Fusion* **125**, 103479 (2026).
69. Han, Z., Yang, F., Huang, J., Zhang, C. & Yao, J. Multimodal dynamics: dynamical fusion for trustworthy multimodal classification. In *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)* 20675–20685 https://doi.org/10.1109/CVPR52688.2022.02005 (2022).
70. Lopez, R., Regier, J., Cole, M. B., Jordan, M. I. & Yosef, N. Deep generative modeling for single-cell transcriptomics. *Nat. Methods* **15**, 1053–1058 (2018).
71. Tang, Z. et al. scTransMIL bridges patient-level disease states and single-cell transcriptomics for cancer screening and heterogeneity inference. *Zenodo* https://doi.org/10.5281/zenodo.21151067 (2026).

Acknowledgements

We thank all members of the Tencent AI for Life Sciences Lab for helpful discussions and constructive feedback. We also acknowledge the researchers and consortia who generated and made publicly available the single-cell datasets used in this study.

Author contributions

F.Y. and J.Y. conceived and designed the project. Z.T. and F.W. developed the algorithm and conducted experiments under the supervision of J.Y. Z.T. and F.W. analyzed the results. Y.L., Jia.Z. (Jiale Zhou), Y.S., S.C., and Jun.Z. (Jun Zhu) assisted with data analysis, validated the algorithm, and provided critical feedback. Z.T. and F.W. wrote the manuscript. Z.T. finished the figures under the guidance of F.W. and J.Y. J.Y., J.S., L.Y., and C.Y.-C.C. gave suggestions for the manuscript. F.W., F.Y., and L.Y. gave suggestions for the project. Z.T. and F.W. contributed equally. All authors reviewed and approved the manuscript.

Funding

This work was supported by the National Natural Science Foundation of China (Grant No. 62541160275, to C.Y.-C.C.) and the Prevention and Control of Emerging and Major Infectious Diseases-National Science and Technology Major Project (No. 2025ZD01901502, to C.Y.-C.C.).

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41467-026-77538-5.

Correspondence and requests for materials should be addressed to Fang Wang, Linlin You, Calvin Yu-Chian Chen or Jianhua Yao.

Peer review information *Nature Communications* thanks Yoshiaki Tanaka and the other anonymous reviewers for their contribution to the peer review of this work. A peer review file is available.

Reprints and permissions information is available at http://www.nature.com/reprints

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

© The Author(s) 2026